

CENG381 Stochastic Processes

Random Walks on Graphs - Question Set 1

Q1 (30 points). Consider a simple random walk on a graph $G = (V, E)$. At each step, if the current state is vertex x , the chain moves to a uniformly chosen neighbour of x .

- Write a formula for the transition probability $P(x, y)$ in terms of the degree $\deg(x)$ of vertex x .
 - Show that the stationary distribution of this random walk is $\pi(x) = \deg(x) / (2|E|)$, where $|E|$ is the number of edges.
 - Consider the graph below (a 'star' with hub H and leaves A, B, C, D): H is connected to A, B, C, D ; there are no other edges. Write the transition matrix and find π for each vertex.
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Q2 (35 points). The random walk on the n -cycle has state space $Z_n = \{0, 1, \dots, n-1\}$ with transition probabilities:

$$\begin{aligned} P(j, k) &= 1/2 & \text{if } k &= j+1 \pmod n \\ P(j, k) &= 1/2 & \text{if } k &= j-1 \pmod n \\ P(j, k) &= 0 & \text{otherwise} \end{aligned}$$

- For $n = 5$ (5-cycle), write the full 5×5 transition matrix P .
 - Verify that the uniform distribution $\pi(j) = 1/5$ for all j is stationary by checking $\pi * P = \pi$.
 - What is the period of any state in the 5-cycle? Is the period the same for the 4-cycle? Explain the difference between odd and even cycles.
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Q3 (35 points). A random mapping representation of a Markov chain with transition matrix P on state space X is a function $f: X \times \Lambda \rightarrow X$ together with a Λ -valued random variable Z , satisfying $P\{f(x, Z) = y\} = P(x, y)$ for all x, y .

- For the simple random walk on the 3-cycle (states $\{0, 1, 2\}$), let $\Lambda = \{-1, +1\}$ with Z equally likely to be -1 or $+1$. Define $f(x, z) = (x + z) \pmod 3$. Verify that this is a valid random mapping representation.
- Suppose $Z_1, Z_2, \dots$ are i.i.d. copies of Z and X_0 has some distribution μ . Define $X_n = f(X_{n-1}, Z_n)$. Show that $(X_0, X_1, \dots)$ is a Markov chain with transition matrix P and initial distribution μ .
- For the general n -cycle, propose a random mapping representation using Z uniform on $\{-1, +1\}$ and $f(x, z) = (x + z) \pmod n$. What is $P(j, k)$ under this representation?